Problem 1. For each of the following algorithms,

(i) Determine what condition(s) must be true for the best case to occur.

(ii) Determine what condition(s) must be true for the worst case to occur.

(iii) Determine the expected case running time. Fully justify your solution.

Assume that $\texttt{Random}(n)$ takes constant time and returns a number between 1 and n uniformly randomly.

(a)

```
function Func(A[],n)
    x = 0
    for i = 1 to n do
        k = Random(i^2)
        for j = 1 to k do
            x = x + i - j
        end for
    end for
    return x
end function
```

$T_i(n)$ $\quad$ ck

Loop that runs a random # of times

BC: $k=1$ ALWAYS

WC: $k=i^2$ ALWAYS

EC: $ET_i(n) = \sum_{q=1}^{i^2} Pr(k=q)\, t(k=q)$ $\quad$ $1/i^2$ $\quad$ cq

$$ET_i(n) = \sum_{q=1}^{i^2} \frac{1}{i^2} \cdot cq$$

$$ET_i(n) = \frac{c}{i^2} \sum_{q=1}^{i^2} q$$

$$= \frac{c}{i^2} \cdot \frac{i^2(i^2+1)}{2}$$

$$= \frac{c(i^2+1)}{2}$$

$\Theta(i^2)$

$$T(n) = \sum_{i=1}^{n} T_i(n)$$

$$ET(n) = E\left(\sum_{i=1}^{n} T_i(n)\right)$$

$$= \sum_{i=1}^{n} ET_i(n)$$

$$= \sum_{i=1}^{n} ci^2$$

$$= c\,\frac{n(n+1)(2n+1)}{6}$$

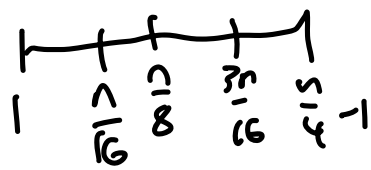
$\Theta(n^3)$.

(b)

```
function Func(A[],n)
    if n ≤ 10 then
        return 1
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + A[i] − A[j]
        end for
    end for
    k = Random(⌊n/4⌋ − 1)
    x = x + Func(A[], 4k)
    x = x + Func(A[], n − 4k)
    return x
end function
```

$Pr(\ )\ ET$ $\qquad$ $Pr(\ )\ ET(\ )$

$\frac{1}{4}(\ )$ $\qquad$ $\frac{1}{2}$



ANT 1 Sol'n.

$1 \le k < \frac{n}{16}$

$\frac{n}{16} \le k < \frac{3n}{16}$

$\frac{3n}{16} \le k$

$$Pr(k \le y) = y/n$$

(c)

```
function Func(A[],n)
    x = 0
    repeat
        k = Random(n)
        for i = 1 to n^2 do
            x = x + A[1 + (ij^2 mod n)]
        end for
    until k > n/2
    return x
end function
```

cn^2

BC: $k > \frac{n}{2}$ first time n^2

WC: $k \le \frac{n}{2}$ always $O(\infty)$

EC: X = # of times $k \le \frac{n}{2}$

$$T(n) = cn^2 + X \cdot cn^2$$

$$ET(n) = cn^2 + cn^2 E(X)$$

$$ET(n) = cn^2 + cn^2 \underbrace{\sum_{i=1}^{\infty} (1/2)^i}_{\Theta(1)}$$

cuz $r < 1$

$$E(X) = \sum_{i=1}^{\infty} Pr(X \ge i)$$

$$Pr(k \le n/2) = \frac{n/2}{n} = 1/2$$

$$Pr(X \ge 1) = 1/2$$

$$Pr(X \ge 2) = (1/2)^2$$

$$Pr(X \ge i) = (1/2)^i$$

$$ET(n) \in \Theta(n^2)$$

what if

$$\sum_{i=0}^{\log_6(n)} 2^i \in \Theta(2^{\log_6(n)})$$

$$\Theta(n^{\log_6(2)})$$

0, 1, 2

(d)

```
function Func(A[],n)
    if n ≤ 1 then
        return 1
    end if
    x = 0
    for i = 1 to n do
        x = x + A[i] − A[j]
    end for
    k = Random(n)
    if k mod 3 = 2 then
        x = x + Func(A[], n − 2)
        x = x + Func(A[], n − 4)
    else
        x = x + Func(A[], n/6)
    end if
    return x
end function
```

$\frac{1}{3}$

$\frac{2}{3}$

BC: $k \bmod 3 \neq 2$ always

WC: $k \bmod 3 = 2$ always

EC:

$$ET(n) = cn + \frac{1}{3}\left(ET(n-2) + ET(n-4)\right) + \frac{2}{3} ET(n/6)$$

$\geq cn.$

$$ET(n) \leq cn + \frac{1}{3}\left(ET(n) + ET(n)\right) + \frac{2}{3} ET\left(\frac{n}{6}\right)$$

$$ET(n) \leq cn + \frac{2}{3} ET(n) + \frac{2}{3} ET(n/6)$$

$$\frac{1}{3} ET(n) \leq cn + \frac{2}{3} ET(n/6)$$

$$ET(n) \leq \alpha n + 2ET(n/6)$$

$$ET(n/6) \leq \alpha \cdot \frac{n}{6} + 2ET\left(\frac{n/6}{6}\right)$$

$$ET(n) \leq \alpha n + 2\left[\alpha \cdot \frac{n}{6} + 2ET\left(\frac{n/6}{6}\right)\right]$$

$$ET(n) \leq \alpha n + \frac{\alpha n}{3} + 4ET(n/36)$$

In general,

$$ET(n) \leq \sum_{i=0}^{\gamma-1} \frac{\alpha n}{3^i} + 2^{\gamma} ET(n/6^{\gamma})$$

$$\frac{n}{6^{\gamma}} = 1$$

$$n = 6^{\gamma}$$

$$\gamma = \log_6(n)$$

$$ET(n) \leq \alpha n \underbrace{\sum_{i=0}^{\gamma-1} (1/3)^i}_{\Theta(1) \text{ cuz } r<1} + \underbrace{2^{\log_6(n)}}_{\Theta(n^{\log_6(2)})} \cancel{ET(1)}^{\,c}$$

$O(n)$

$\Theta(n)$ $\quad$ $n^{\log_6(2)}$

Thus $ET(n) \in \Theta(n)$.